Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following molecule(s) is/are showing optical isomerism?

(A) $[\mathrm{Co}\,(\mathrm{NH_3})_3\,l]^{\oplus}$

(B) $[\mathrm{Co(en)}\,(\mathrm{NH_3})_2]^{2+}$

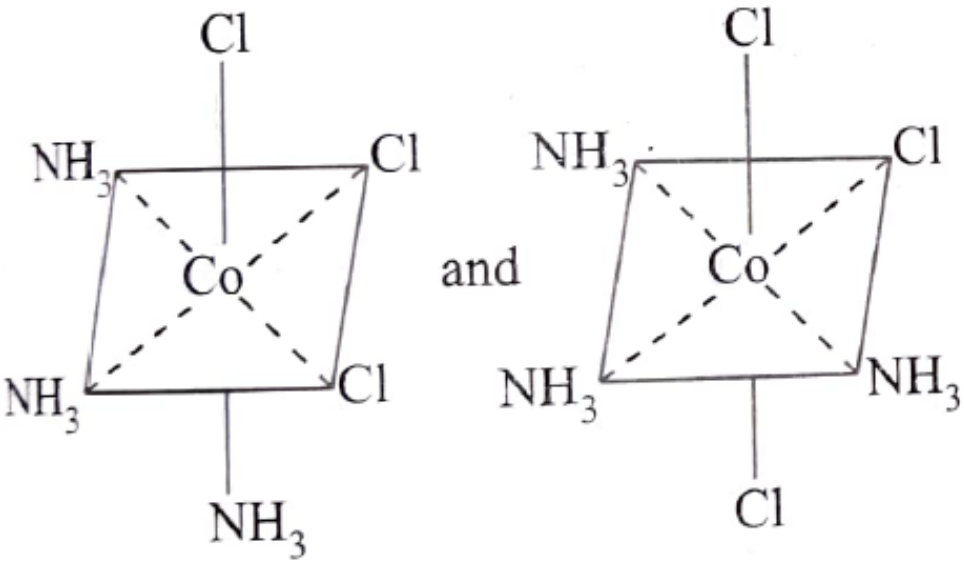
(C) $[\mathrm{Co}\,(\mathrm{H_2O})_4\,(\mathrm{en})]^{3+}$

(D) $[\mathrm{Co(en)_2}\,(\mathrm{NH_3})_2]^{3+}$

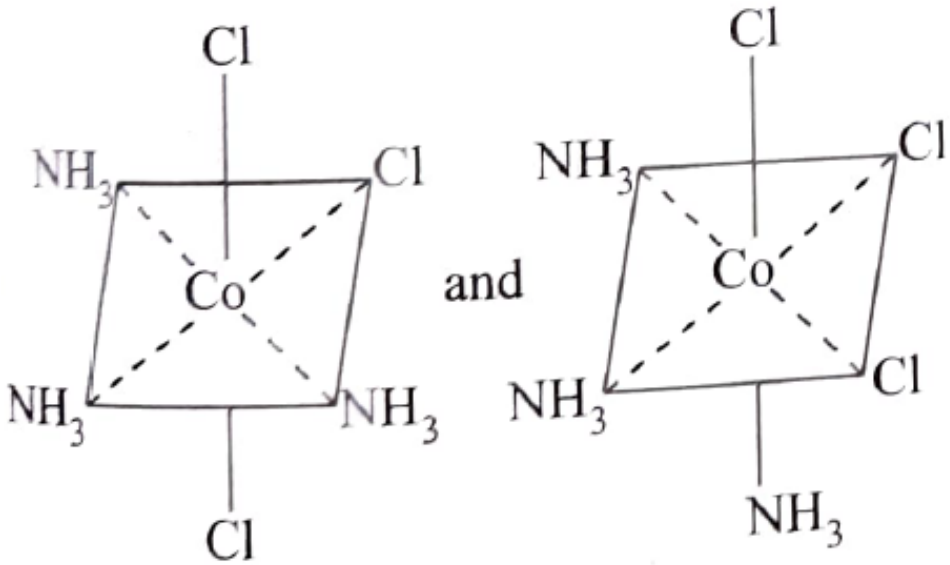
Q.2

Which of the following pairs of structures represent facial and meridional isomers (geometrical isomers) respectively?

(A)

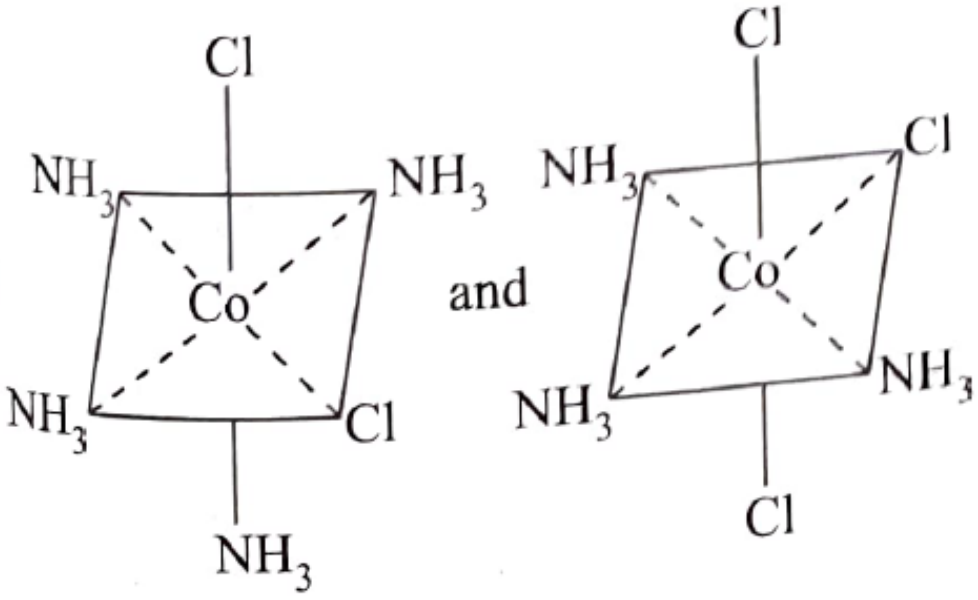


(B)

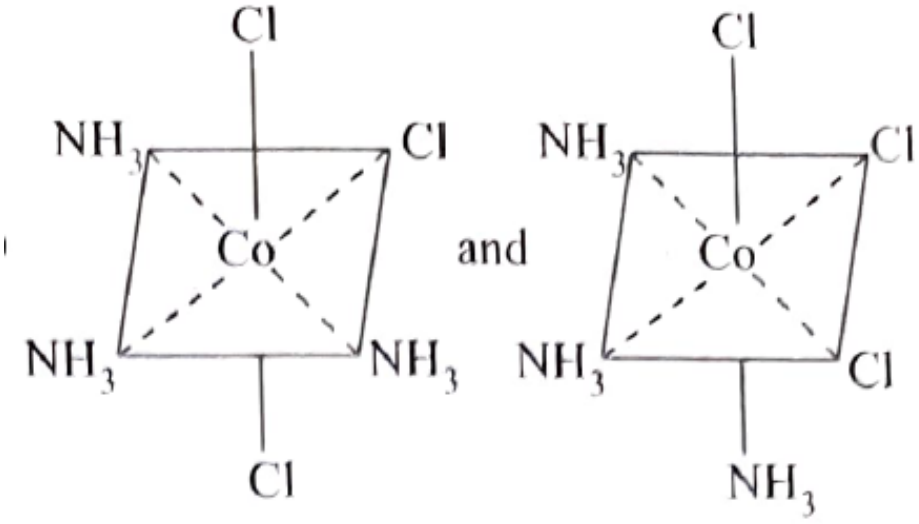


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(C)



(D)



Q.3

$[Co\,(NH_3)_5\,Br]\,SO_4$ and $[Co\,(NH_3)_5\,SO_4]\,Br$ are example of which type of isomerism

(A) Linkage

(B) Geometrical

(C) Ionization

(D) Optical

Q.4

A solution containing 2.675 g of $CoCl_3.6NH_3$ (molar mass $= 267.5$ g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 4.78 g of $AgCl$ (molar mass $= 143.5$ g mol^{-1}). The formula of the complex is (At. mass of $Ag = 108u$)

(A) $[Co\,(NH_3)_6]\,Cl_3$

(B) $[CoCl_2\,(NH_3)_4]\,Cl$

(C) $[CoCl_3\,(NH_3)_3]$

(D) $[CoCl\,(NH_3)_5]\,Cl_2$

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Q.5

The type of isomerism present in Pentaminenitrochromium (III) chloride is

(A) optical

(B) linkage

(C) ionisation

(D) polymerisation.

Q.6

Which one of the following has largest number of isomers?

(A) $[\mathrm{Ir}\,(\mathrm{PR_3})_2\,\mathrm{H(CO)}]^{2+}$

(B) $[\mathrm{Co}\,(\mathrm{NH_3})_5\,\mathrm{Cl}]^{2+}$

(C) $[\mathrm{Ru}\,(\mathrm{NH_3})_4\,\mathrm{Cl_2}]^{+}$

(D) $[\mathrm{Co(en)_2Cl_2}]^{+}$

Q.7

Consider the following complexes.
$[\mathrm{MA_6}]$, $[\mathrm{MA_5\ B}]$, $[\mathrm{MA_2\ B_4}]$, $[\mathrm{MA_3\ B_3}]$ $[\mathrm{MA_4\ B_2}]$
Find the number of complexes will show geometrical isomerism.

(A) 1

(B) 3

(C) 4

(D) 2

Q.8

Total number of geometrical isomers from the complex
$[\mathrm{RhCl(CO)\,(PPh_3)\,(NH_3)}]$ is

(A) 4

(B) 2

(C) 3

(D) 1

Q.9

Which of the following complex will show geometrical as well as optical isomerism (en=ethylenediammine)

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(A) $[Pt\,(NH_3)_2\,Cl_2]$

(B) $[Pt(en)Cl_4]$

(C) $[Pt(en)_3]^{4+}$

(D) $[Pt(en)_2Cl_2]$

Q.10

The total number of possible isomers of the complex compound $[Cu^{II}\,(NH_3)_4]\,[Pt^{II}Cl_4]$ is

(A) 3

(B) 6

(C) 5

(D) 4

Q.11

Which of the following will give maximum number of isomers?

(A) $[Co\,(NH_3)_4\,Cl_2]$

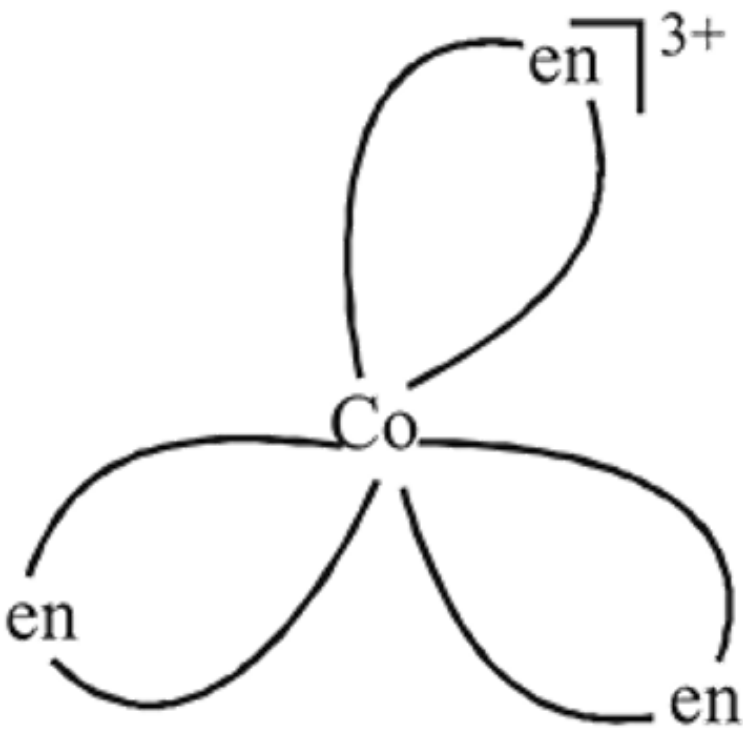
(B) $[Ni(en)\,(NH_3)_4]^{2+}$

(C) $[Ni\,(C_2O_4)\,(en)_2]$

(D) $[Cr(SCN)_2\,(NH_3)_4]^{2+}$

Q.12

The complex given is



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(i) non-superimposable on its mirror images
(ii) optically active
(iii) rotate plane polarised light
(iv) planar

(A) (i) and (ii)

(B) (i) and (iv)

(C) (i), (ii) and (iii)

(D) (ii) only

Q.13

$[\mathrm{Co}\left(\mathrm{NH_3}\right)_4\left(\mathrm{NO_2}\right)_2]\,\mathrm{Cl}$ exhibits

(A) linkage isomerism, ionization isomerism and geometrical isomerism

(B) ionization isomerism, geometrical isomerism and optical isomerism

(C) linkage isomerism, geometrical isomerism and optical isomerism

(D) linkage isomerism, ionization isomerism and optical isomerism

Q.14

Coordination number of Cr is six. A complex with $\mathrm{C_2O_4^{2-}}$, en and superoxide $\mathrm{O_2^-}$ will be in the ratio to make complex $[\mathrm{Cr}\left(\mathrm{C_2O_4}\right)_{\mathrm{x}}\left(\mathrm{en}\right)_{\mathrm{y}}\left(\mathrm{O_2}\right)_{\mathrm{z}}]^-$:

(A) $x=1, y=1, z=1$

(B) $\mathrm{x}=1, \mathrm{y}=1, \mathrm{z}=2$

(C) $x=1, y=2, z=2$

(D) $x=2, y=1, z=1$

Q.15

For complex ion/compound formation reactions
(I) $\mathrm{Co^{3+}(aq)+EDTA^{4-}\longrightarrow P}$
(II) $\mathrm{Ni^{2+}(aq)+dmg(\text{ excess })\overset{NH_4OH}{\longrightarrow}Q}$
(III) $\mathrm{Zn^{2+}(aq)+\text{ gly }(\text{ excess })\longrightarrow R}$
(IV) $\mathrm{Pt^{4+}(aq)+\text{ en }(\text{ excess })\longrightarrow S}$
which of the following complex ion/compound does not exhibit optical activity?

(A) P

(B) Q

(C) R

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(D) S

Q.16

An octahedral complex with molecular composition $M.5NH_3 \cdot Cl \cdot SO_4$ has two isomers,
A and B. The solution of A gives a white precipitate with $AgNO_3$ solution and the solution of B gives white precipitate with $BaCl_2$ solution. The type of isomerism exhibited by the complex is:

(A) Linkage isomerism

(B) Ionisation isomerism

(C) Coordinate isomerism

(D) Geometrical isomerism

Q.17

Type of isomerism which exists between $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ is :

(A) Linkage isomerism

(B) Coordination isomerism

(C) Ionisation isomerism

(D) Solvate isomerism

Q.18

Which of the following complex ions will exhibit optical isomerism? $(en = 1,2$ -diamine ethane $)$.

(A) $[Cr(NH_3)_2Cl_2]^+$

(B) $[Co(en)_2Cl_2]^+$

(C) $[Co(NH_3)_4Cl_2]^+$

(D) $[Zn(en)_2]^{2+}$

Q.19

The complex ion $[Pt(NO_2)(Py)(NH_3)(NH_2OH)]$
will give

(A) 2 isomers (Geometrical)

(B) 3 isomers (Geometrical)

(C) 6 isomers (Geometrical)

(D) 4 isomers (Geometrical)

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Q.20

$[\mathrm{Ni}\,(\mathrm{NH_3})_6]^{2+} + \mathrm{en} \longrightarrow \mathrm{X} + 2\mathrm{NH_3}$
$[\mathrm{Ni}\,(\mathrm{NH_3})_6] + 2\mathrm{en} \longrightarrow \mathrm{Y} + 4\mathrm{NH_3}$
$[\mathrm{Ni}\,(\mathrm{NH_3})_6 + 3\mathrm{en} \longrightarrow \mathrm{Z} + 6\mathrm{NH_3}$

Which of them show optical as well as geometrical isomerism?

(A) X

(B) Y

(C) Z

(D) All of these

Answers & Solutions

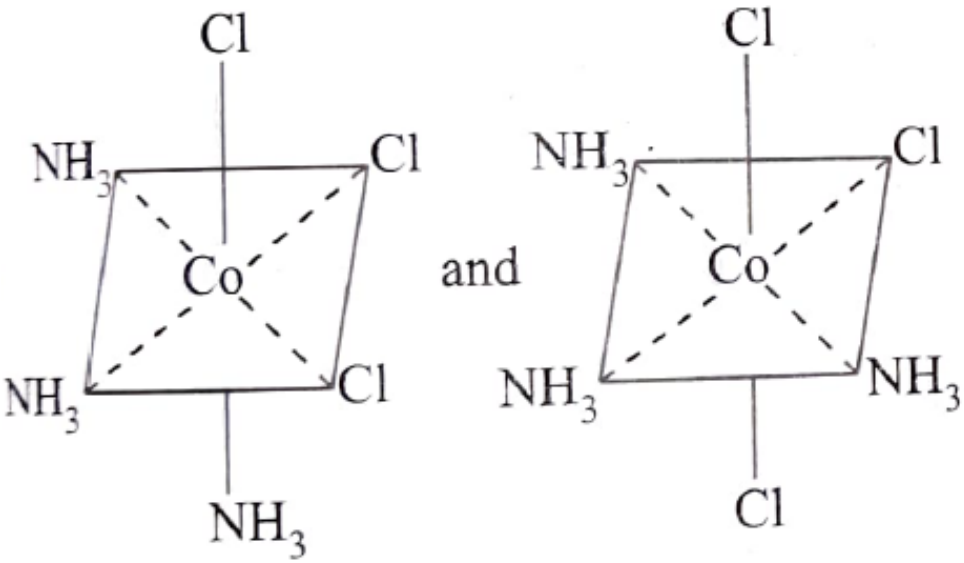
Q.1 Answer:

$[\mathrm{Co(en)_2}\,(\mathrm{NH_3})_2]^{3+}$

Solution:

Optical isomerism is shown by:
(i) complexes of the type $[M(AA)X_2Y_2]$ containing one symmetrical bidentate ligand.
(ii) complexes of the type $[M(AA)_3]$
containing a symmetrical bidentate ligand.
(iii) complexes of the type $[M(AA)_2X_2]$
So, from the given option, only $[\mathrm{Co(en)_2}\,(\mathrm{NH_3})_2]^{3+}$
Shows optical isomerism and the others doesn't.

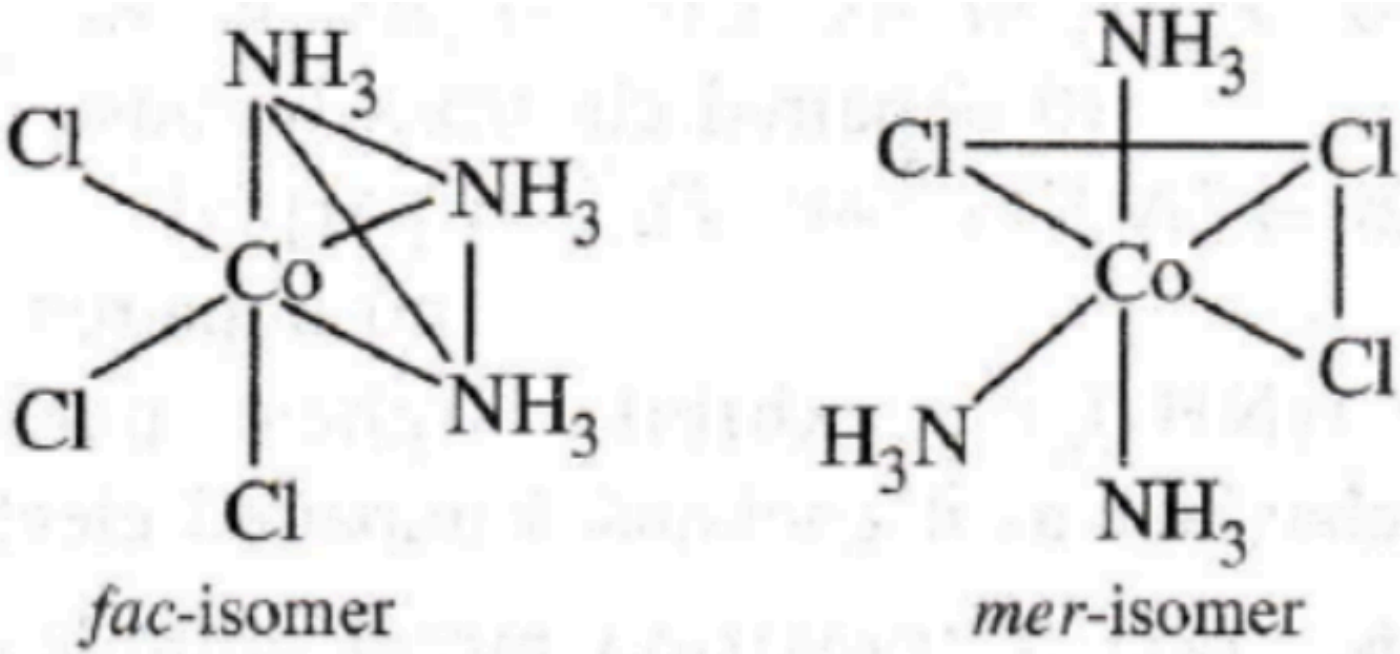
Q.2 Answer:



Solution:

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$[Co(NH_3)_3Cl_3]$ **exhibits fac-mer isomerism.**



Q.3 Answer:

Ionization

Solution:

The two given compounds have same composition but in solution both will give different ions. This type of isomerism is known as ionisation isomerism.

Q.4 Answer:

$[Co(NH_3)_6]Cl_3$

Solution:

$CoCl_3 \cdot 6NH_3 \longrightarrow xCl^-$

$2.675\ g$

$xCl^- + AgNO_3 \longrightarrow xAgCl\downarrow$

$4.78\ g$

Number of moles of the complex

$= \frac{2.675}{267.5} = 0.01$ **moles**

Number of moles of $AgCl$ **obtained**

$= \frac{4.78}{143.5} = 0.03$ **moles**

$\therefore$ **No. of moles of** $AgCl$ **obtained** $= 3\times$ **No. of moles of complex**

$\therefore n = \frac{0.03}{0.01} = 3$

Q.5 Answer:

linkage

Solution:

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The chemical formula of Pentaminenitrochromium (III) chloride is

$[Cr(NH_3)_5NO_2]Cl_2$

It can exist in following two structures

$[Cr(NH_3)_5NO_2]Cl_2$ **and**

$[Cr(NH_3)_5ONO]Cl_2$

Therefore the type of isomerism found in this compound is linkage isomerism as nitro group is linked through N **as** $-NO_2$ **or through** O **as** $-ONO$.

Q.6 Answer:

$[Co(en)_2Cl_2]^+$

Solution:

Isomers of given complexes are as follows

$[Ru(NH_3)_4Cl_2]^+$, $[Co(NH_3)_5Cl]^{2+}$

cis and trans none

$[Ir(PR_3)_2H(CO)]^{2+}$, $[Co(en)_2Cl_2]^+$

cis and trans cis and trans and optical isomers

Thus $[Co(en)_2Cl_2]^+$ **has large number of isomers.**

Q.7 Answer:

3

Solution:

$[MA_6]$ **No. of geometrical isomers** $[MA_5B]$ **No. of geometrical isomers** $\left.\begin{matrix}[MA_2B_4]\\[MA_3B_3]\\[MA_4B_2]\end{matrix}\right\}$ **Each**

show 2 geometical isomers

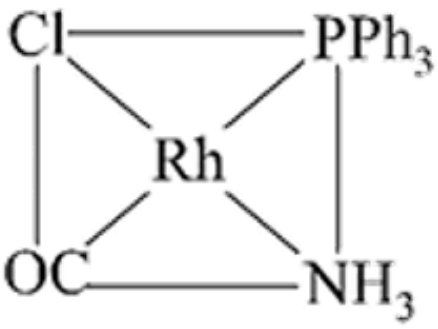
Thus, 3 complexes will show geometrical isomerism.

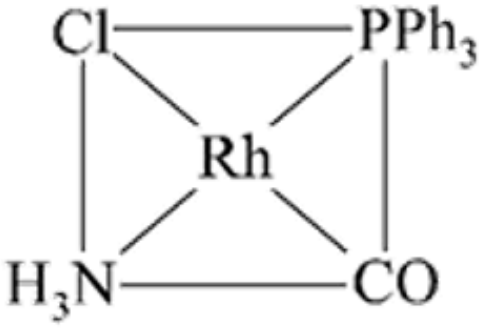
Q.8 Answer:

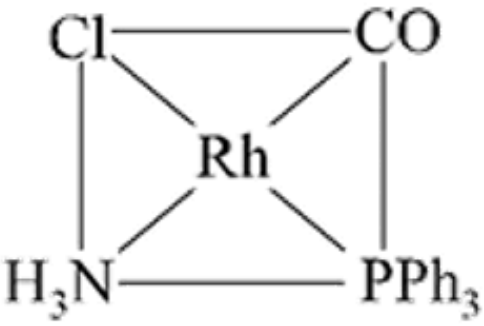
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Solution:

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The number of geometrical isomers is 3 .

Q.9 Answer:

$[Pt(en)_2Cl_2]$

Solution:

$[Pt(en)_2Cl_2]$ **is a complex of the type** $[M(AA)_2\ B_2]$ **which is octahedral. Such compounds exhibit optical and geometrical isomerism both.**

Q.10 Answer:

4

Solution:

For $\left[Cu^{II}\ (NH_3)_4\right]\ \left[Pt^{I}Cl_4\right]$ **four isomers are possible which are** $[Cu\ (NH_3)_4]\ [PtCl_4]$ $[CuCl_4]\ [Pt\ (NH_3)_4]\ [PtCl_3\ (NH_3)]\ [Cu\ (NH_3)_3\ Cl]$ **and** $[Pt\ (NH_3)_3\ Cl]\ [Cu\ (NH_3)\ Cl_3]$

Q.11 Answer:

$[Cr(SCN)_2\ (NH_3)_4]^{2+}$

Solution:

The complex ion $[Cr(SCN)_2\ (NH_3)_4]^{2+}$ **can exhibit geometrical and linkage isomerism**

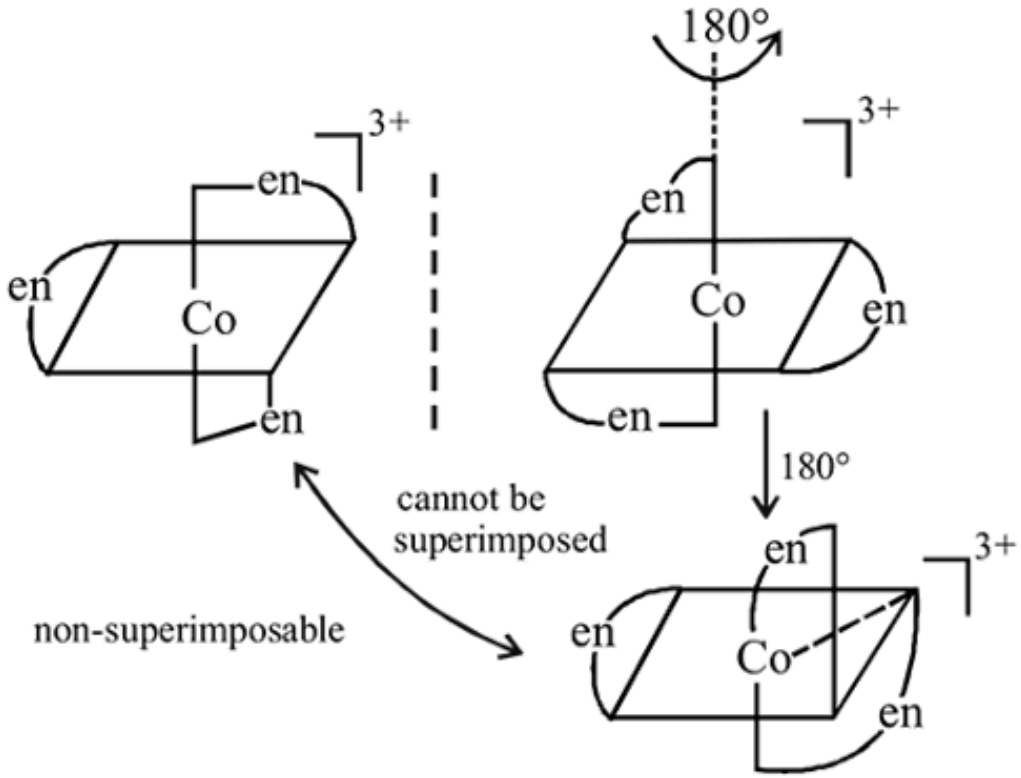
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Q.12 Answer:

(i), (ii) and (iii)

Solution:

Complex is not superimposable on its mirror image hence optically active i.e., rotate plane polarized light.



Q.13 Answer:

linkage isomerism, ionization isomerism and geometrical isomerism

Solution:

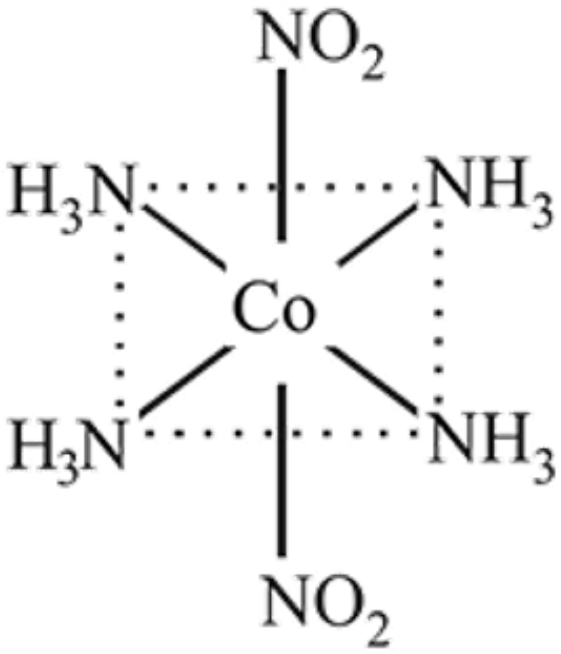
The given compound may have linkage isomerism due to presence of NO_2 group which may be in the form $-NO_2$ or -ONO. $[Co(NH_3)_4(NO_2)_2]Cl$ & $[Co(NH_3)_4(ONO)_2]Cl$

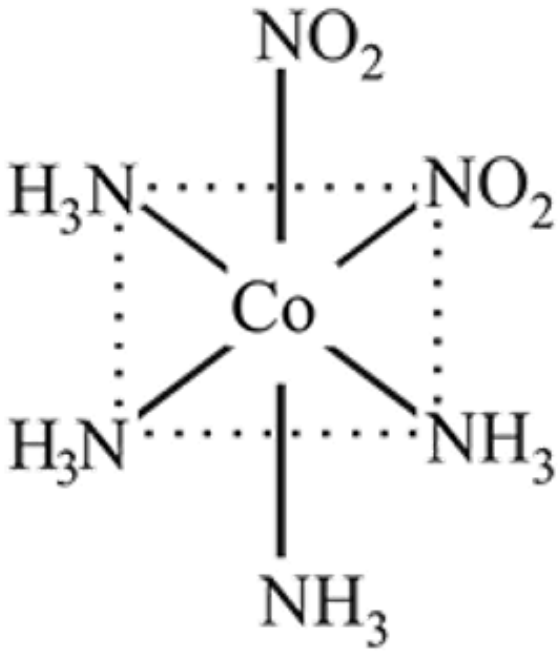
It may have ionisation isomerism due to presence of two ionisable groups $-NO_2$ & $-Cl$

$[Co(NH_3)_4Cl(NO_2)]NO_2$ & $[Co(NH_3)(NO_2)_2]Cl$

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It may have geometrical isomerism in the form of cis trans form as follows:





Q.14 Answer:

$x = 1, y = 1, z = 2$

Solution:

As co-ordination of Cr is 6
$C_2O_4^{2-}$ & en are bidentate ligands while O_2^- is unidentate
so $x = 1, y = 1, z = 2$
$[Cr\,(C_2O_4)_1\,(en)_1\,(O_2)_2]^-$

Q.15 Answer:

Q

Solution:

$Ni^{2+}(aq)$ with DMG (dimethylglyoxime) form square planar complex which is obtically inactive.
While $[Co(EDTA)]^-$, $[Zn(gly)_2]^{2-}$
and $[Pt(en)_3]^{4+}$ are optically active as they do not have P.O.S.

Q.16 Answer:

Ionisation isomerism

Solution:

The two possible isomers for the given octahedral complex are $[M\,(NH_3)_5\,SO_4]\,Cl$ and $[M\,(NH_3)_5\,Cl]\,SO_4$. They respectively give chloride ion (indicated by precipitation with $BaCl_2$) and SO_4 ion (indicated by precipitation with $AgNO_3$). Hence the type of isomerism exhibited by the complex is ionization isomerism.

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Q.17 Answer:

Linkage isomerism

Solution:

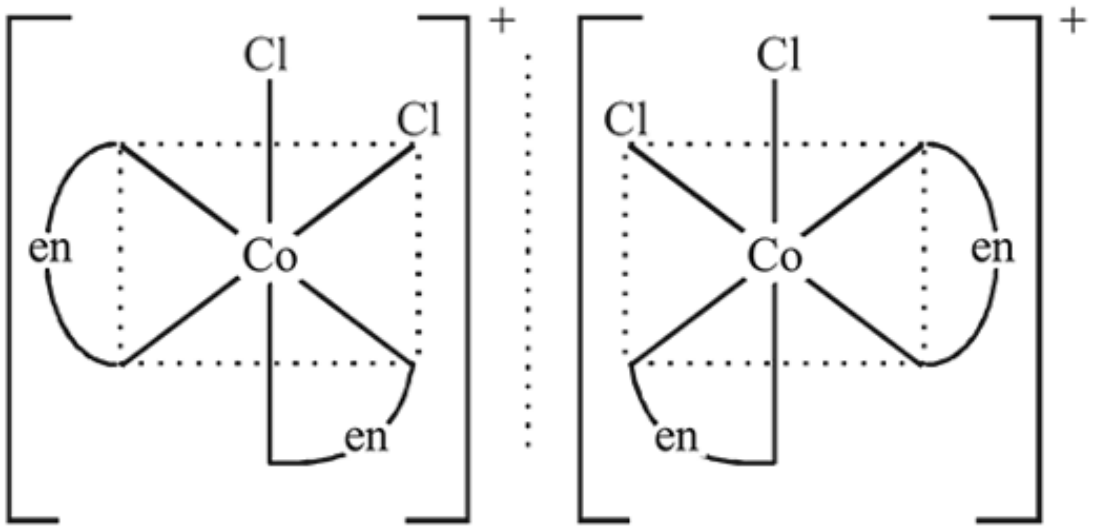
The compound shows linkage isomerism because the ligand in the compound is an ambidenate ligand that can bond at more than one atomic site. i.e., : NCS^- **and :** SCN^-

Q.18 Answer:

$[Co(en)_2Cl_2]^+$

Solution:

Non superimposable mirror images are called optical isomers and may be described as "chiral'. They are also called enantiomers and rotate plane polarised light in opposite directions.

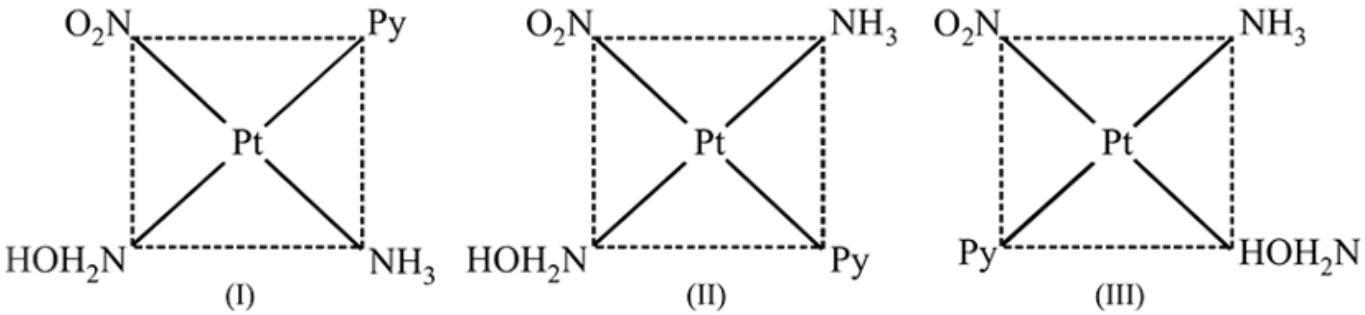


Q.19 Answer:

3 isomers (Geometrical)

Solution:

Complexes of the type M_{ABCD} **may exist in three isomeric form. Similarily** $[Pt(NO_2)(Py)(NH_3)(NH_2OH)^+]$ **may exist in three isomeric form.**



Q.20 Answer:

Y

Solution:

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$X = [Ni(en)(NH_3)_4]^{2+}$ **neither G.I. nor OI**
$Y = [Ni(en)_2(NH_3)_2]^{2+}$ **G. I as well as** OI
$Z = [Ni(en)_3]^{2+}$ **does not show G.I. while show Ol**

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